# Contracts to Effects Compiler Typesoundness

## Overview

This part shows how the compiler remains typesoundness of the source program to the target program. Since the compiler introduces effects and explicitly declares it everywhere (in lambdas, and perform operations), the function type is augmented with the effect. The compiler also does not introduce any handler, this makes the whole program (potentially) produce the effect as well.

The environment that will be used to type check target programs has the following $\Sigma$ signature.

$$\Sigma(\mathtt{h}_{\mathscr{C}}) = \left\{\mathscr{C} : \forall\alpha.\langle \mathtt{blab}, \langle \alpha \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, \alpha\rangle\rangle \to \alpha\right\}$$

## Type Translation

$\cdot^* : \text{Contracts } \tau \to \text{Effects } \tau$

$$\begin{aligned}
\mathsf{num}^* &= \mathsf{num} \\
\mathsf{bool}^* &= \mathsf{bool} \\
\mathsf{unit}^* &= \mathsf{unit} \\
\langle \tau_1, \tau_2 \rangle^* &= \langle \tau_1^*, \tau_2^* \rangle \\
(\tau_1 \to \tau_2)^* &= \tau_1^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau_2^* \\
(x_1 : \tau_1, ..., x_n : \tau_n)^* &= x_1 : \tau_1^*, ..., x_n : \tau_n^*
\end{aligned}$$

## Compiler Type Preservation

**Theorem 0.1.** (Compiler Type Preservation)

If $\Gamma \vdash e : \tau \rightarrowtail e'$ then $\Gamma^* \vdash e' : \tau^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$.

*Proof.*

By induction on structure of $\Gamma \vdash e : \tau \rightarrowtail e'$.

- Case $e = x$, $e = n$, $e = b$, $e = \mathsf{unit}$

  $e' = e$ — Given

  $\Gamma^* \vdash_{\mathrm{val}} e' : \tau^*$ — T-E-Var, T-Num, T-Bool, T-Unit

  $\Gamma^* \vdash e' : \tau^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — T-E-Val

- Case

$$\frac{\Gamma, x : \tau_1 \vdash e : \tau_2 \rightarrowtail e''}{\Gamma \vdash \lambda x : \tau_1.e : \tau_1 \to \tau_2 \rightarrowtail \lambda x : \tau_1^*.e''}$$

$\Gamma^*, x : \tau_1^* \vdash e'' : \tau_2^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
$\Gamma^* \vdash_{\text{val}} \lambda x : \tau_1^*.e'' : \tau_1^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau_2^*$ `T-E-Abs`
$\Gamma^* \vdash_{\text{val}} \lambda x : \tau_1^*.e'' : (\tau_1 \to \tau_2)^*$ Type compilation $\cdot^*$
$\Gamma^* \vdash \lambda x : \tau_1^*.e'' : (\tau_1 \to \tau_2)^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ `T-E-Val`

- Case

$$\frac{\Gamma \vdash e_1 : \tau_1 \to \tau_2 \rightarrowtail e_1' \qquad \Gamma \vdash e_2 : \tau_1 \rightarrowtail e_2'}{\Gamma \vdash e_1\ e_2 : \tau_2 \rightarrowtail e_1'\ e_2'}$$

$\Gamma^* \vdash e_1' : \tau_1^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau_2^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
$\Gamma^* \vdash e_2' : \tau_1^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
$\Gamma^* \vdash e_1'\ e_2' : \tau_2^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ `T-E-App`

The effect on the arrow and the effect of the premises are both $\langle \mathtt{h}_{\mathscr{C}} \rangle$.

- Case $\mu x : \tau.e$, $\langle e_1, e_2 \rangle$, $\mathsf{inj.l}(e)$, $\mathsf{inj.r}(e)$, if $e_1$ then $e_2$ else $e_3$, $o(e_1, ..., e_n)$

$\Gamma^* \vdash e_i' : \tau_i^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ for every sub-expression I.H.
$\Gamma^* \vdash e' : \tau^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ lifted rule of Base

For $\mu x : \tau.e$, the binder is $x : \tau^*$, and the I.H. is used under $\Gamma^*, x : \tau^*$.

- Case

$$\frac{\Gamma \vdash e_3 : \tau \to \mathsf{bool} \rightarrowtail e_5 \qquad \Gamma \vdash e_4 : \tau \rightarrowtail e_6}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\mathsf{flat}(e_3), e_4) : \tau \rightarrowtail \mathtt{perform}\ \mathscr{C}\ \tau^*\ \langle k, \langle e_5, e_6 \rangle \rangle}$$

$\Gamma^* \vdash e_5 : \tau^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool} \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
$\Gamma^* \vdash e_6 : \tau^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
$\Gamma^* \vdash \mathtt{perform}\ \mathscr{C}\ \tau^*\ \langle k, \langle e_5, e_6 \rangle \rangle : \tau^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ Lemma 0.2

This is the only rule that introduces $\mathtt{h}_{\mathscr{C}}$.

- Case

$$\frac{\Gamma \vdash e_1 : \tau_1 \to \tau_2 \rightarrowtail e_2 \qquad \Gamma \vdash \lambda f : \tau_1 \to \tau_2.\lambda x : \tau_1.\mathsf{mon}_j^{k,l}\Big(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail e_3}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1) : \tau_1 \to \tau_2 \rightarrowtail e_3\ e_2}$$

$\Gamma^* \vdash e_2 : (\tau_1 \to \tau_2)^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
$\Gamma^* \vdash e_3 : ((\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2))^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
$\Gamma^* \vdash e_3 : (\tau_1 \to \tau_2)^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} (\tau_1 \to \tau_2)^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ def. of $\cdot^*$

$\Gamma^* \vdash e_3\ e_2 : (\tau_1 \to \tau_2)^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ T-E-App

- Case

$$\frac{\Gamma \vdash e_1 : \tau_1 \to \tau_2 \rightarrowtail e_2 \qquad \Gamma \vdash e_w : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail e_3}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e_1) : \tau_1 \to \tau_2 \rightarrowtail e_3\ e_2}$$

where $e_w = \lambda f : \tau_1 \to \tau_2.\lambda x : \tau_1.\mathsf{mon}_j^{k,l}\left(\kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right], f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$

$\Gamma^* \vdash e_2 : (\tau_1 \to \tau_2)^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
$\Gamma^* \vdash e_3 : (\tau_1 \to \tau_2)^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} (\tau_1 \to \tau_2)^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H., def. of $\cdot^*$
$\Gamma^* \vdash e_3\ e_2 : (\tau_1 \to \tau_2)^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ T-E-App

The dependency $y$ is replaced in the source by the indy monitor $\mathsf{mon}_j^{l,j}(\kappa_1, x) : \tau_1$, so the I.H. on $e_w$ already covers $\kappa_2$ in the context $\Gamma, f : \tau_1 \to \tau_2, x : \tau_1$.

- Case

$$\frac{\begin{array}{c} x \notin \mathrm{dom}(\Gamma) \qquad \Gamma, x : \langle \tau_1, \tau_2 \rangle \vdash \mathsf{mon}_j^{k,l}(\kappa_1, \mathsf{inj.l}(x)) : \tau_1 \rightarrowtail e_1 \\ \Gamma, x : \langle \tau_1, \tau_2 \rangle \vdash \mathsf{mon}_j^{k,l}(\kappa_2, \mathsf{inj.r}(x)) : \tau_2 \rightarrowtail e_2 \qquad \Gamma \vdash e : \langle \tau_1, \tau_2 \rangle \rightarrowtail e_3 \end{array}}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\langle \kappa_1, \kappa_2 \rangle, e) : \langle \tau_1, \tau_2 \rangle \rightarrowtail (\lambda x : \langle \tau_1, \tau_2 \rangle^*.\langle e_1, e_2 \rangle)\ e_3}$$

$\Gamma^*, x : \langle \tau_1, \tau_2 \rangle^* \vdash e_1 : \tau_1^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
$\Gamma^*, x : \langle \tau_1, \tau_2 \rangle^* \vdash e_2 : \tau_2^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
$\Gamma^*, x : \langle \tau_1, \tau_2 \rangle^* \vdash \langle e_1, e_2 \rangle : \langle \tau_1, \tau_2 \rangle^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ T-Tuple
$\Gamma^* \vdash_{\mathrm{val}} \lambda x : \langle \tau_1, \tau_2 \rangle^*.\langle e_1, e_2 \rangle : \langle \tau_1, \tau_2 \rangle^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \langle \tau_1, \tau_2 \rangle^*$ T-E-Abs
$\Gamma^* \vdash \lambda x : \langle \tau_1, \tau_2 \rangle^*.\langle e_1, e_2 \rangle : \langle \tau_1, \tau_2 \rangle^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \langle \tau_1, \tau_2 \rangle^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ T-E-Val
$\Gamma^* \vdash e_3 : \langle \tau_1, \tau_2 \rangle^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
$\Gamma^* \vdash (\lambda x.\langle e_1, e_2 \rangle)\ e_3 : \langle \tau_1, \tau_2 \rangle^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ T-E-App

□

**Lemma 0.2.** (Typing a perform of the check effect)

If $\Gamma \vdash e_1 : \tau \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool} \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ and $\Gamma \vdash e_2 : \tau \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ then

$\Gamma \vdash \mathtt{perform}\ \mathscr{C}\ \tau\ \langle k, \langle e_1, e_2 \rangle \rangle : \tau \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$.

*Proof.*

$\Gamma \vdash_{\mathrm{val}} \mathtt{perform}\ \mathscr{C}\ \tau : \langle \mathtt{blab}, \langle \tau \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, \tau \rangle \rangle \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau$ T-E-Perform, $\alpha := \tau, \varepsilon := \langle\rangle$
$\Gamma \vdash \mathtt{perform}\ \mathscr{C}\ \tau : \langle \mathtt{blab}, \langle \tau \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, \tau \rangle \rangle \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ T-E-Val
$\Gamma \vdash k : \mathtt{blab} \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ T-E-Label
$\Gamma \vdash e_1 : \tau \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool} \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ Given
$\Gamma \vdash e_2 : \tau \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ Given
$\Gamma \vdash \langle e_1, e_2 \rangle : \langle \tau \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, \tau \rangle \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ T-Tuple
$\Gamma \vdash \langle k, \langle e_1, e_2 \rangle \rangle : \langle \mathtt{blab}, \langle \tau \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, \tau \rangle \rangle \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ T-Tuple

$\Gamma \vdash \texttt{perform}\ \mathscr{C}\ \tau\ \langle k, \langle e_1, e_2 \rangle\rangle : \tau \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ T-E-App

In the last step, the effect on the arrow and the effect of both premises are the same row $\langle \mathtt{h}_{\mathscr{C}} \rangle$. □

## Wrapped Compiled Program Typing

The deep handler wrapper from the compiler equivalence proof removes $\mathtt{h}_{\mathscr{C}}$ from the effect row. We rewrite the wrapper as follows:

$$\mathsf{wrap} = \mu w : \forall\alpha.\big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\big) \to^{\langle\rangle} \alpha.\Lambda\alpha.\lambda f : \mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha.\ \texttt{handler}\ h_w\ f$$

$$h_w = \big\{\mathscr{C} \to \Lambda\beta.\lambda\langle l, \langle e, v\rangle\rangle.\lambda k : \beta \to^{\langle\rangle} \alpha.\ \text{if}\ w[\mathsf{bool}]\ (\lambda_ : \mathsf{unit}.e\ v)\ \text{then}\ k\ v\ \text{else}\ \texttt{error}_l\big\}$$

Note that $h_w$ signature matches $\Sigma(\mathtt{h}_{\mathscr{C}})$ if we replace $\alpha$ with $\beta$.

**Theorem 0.3.** (Compiler Effect Safety)

If $\cdot \vdash e : \tau \rightarrowtail e'$ then $\cdot \vdash \mathsf{wrap}[\tau^*]\ (\lambda_ : \mathsf{unit}.e') : \tau^* \mid \langle\rangle$.

*Proof.*

| | |
|---|---|
| $\cdot \vdash e' : \tau^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ | Theorem 0.1 |
| $\cdot \vdash_{\mathrm{val}} \lambda_ : \mathsf{unit}.e' : \mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau^*$ | T-E-Abs |
| $\cdot \vdash \mathsf{wrap}[\tau^*] : \big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau^*\big) \to^{\langle\rangle} \tau^* \mid \langle\rangle$ | Lemma 0.4, T-E-Tapp |
| $\cdot \vdash \mathsf{wrap}[\tau^*]\ (\lambda_ : \mathsf{unit}.e') : \tau^* \mid \langle\rangle$ | T-E-App |

□

**Lemma 0.4.** (Type of wrap)

$\cdot \vdash_{\mathrm{val}} \mathsf{wrap} : \forall\alpha.\big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\big) \to^{\langle\rangle} \alpha$

*Proof.*

Assume contexts $\Gamma_w = w : \forall\alpha.\big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\big) \to^{\langle\rangle} \alpha.$

And $\Delta = \Gamma_w, l : \mathtt{blab}, e : \beta \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, v : \beta, k : \beta \to^{\langle\rangle} \alpha.$

| | |
|---|---|
| $\Delta \vdash e\ v : \mathsf{bool} \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ | T-E-App |
| $\Delta \vdash_{\mathrm{val}} \lambda_ : \mathsf{unit}.e\ v : \mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}$ | T-E-Abs |
| $\Delta \vdash w[\mathsf{bool}] : \big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}\big) \to^{\langle\rangle} \mathsf{bool} \mid \langle\rangle$ | T-E-Tapp |
| $\Delta \vdash w[\mathsf{bool}]\ (\lambda_ : \mathsf{unit}.e\ v) : \mathsf{bool} \mid \langle\rangle$ | T-E-App |
| $\Delta \vdash k\ v : \alpha \mid \langle\rangle$ | T-E-App |
| $\Delta \vdash \texttt{error}_l : \alpha \mid \langle\rangle$ | T-E-Error |
| $\Delta \vdash \text{if}\ w[\mathsf{bool}]\ (\lambda_ : \mathsf{unit}.e\ v)\ \text{then}\ k\ v\ \text{else}\ \texttt{error}_l : \alpha \mid \langle\rangle$ | T-If |
| $\Gamma_w \vdash_{\mathrm{val}} h_{w(\mathscr{C})} : \forall\beta.\langle \mathtt{blab}, \langle \beta \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, \beta\rangle\rangle \to^{\langle\rangle} \big(\big(\beta \to^{\langle\rangle} \alpha\big) \to^{\langle\rangle} \alpha\big)$ | T-E-Abs, T-E-TAbs |

$\Gamma_w \vdash_{\text{ops}} h_w : \alpha \mid \langle \mathtt{h}_{\mathscr{C}} \mid \langle\rangle \rangle$ T-E-Ops

$\Gamma_w \vdash_{\text{val}} \mathtt{handler}\ h_w : \big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\big) \to^{\langle\rangle} \alpha$ T-E-Handler

$\Gamma_w, f : \mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha \vdash \mathtt{handler}\ h_w\ f : \alpha \mid \langle\rangle$ T-E-App

$\Gamma_w \vdash_{\text{val}} \lambda f : \mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha.\ \mathtt{handler}\ h_w\ f : \big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\big) \to^{\langle\rangle} \alpha$ T-E-Abs

$\Gamma_w \vdash_{\text{val}} \Lambda\alpha.\lambda f.\ \mathtt{handler}\ h_w\ f : \forall\alpha.\big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\big) \to^{\langle\rangle} \alpha$ T-E-TAbs

$\cdot \vdash_{\text{val}} \mathsf{wrap} : \forall\alpha.\big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\big) \to^{\langle\rangle} \alpha$ T-Fix

□